DD-P-00355A (DSA-DM)
May 27, 1975
Interim Revision of
Fed. Spec. DD-P-355
January 5, 1960

INTERIM FEDERAL SPECIFICATION

PIPET, DROPPING; Glass or Plastic

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification DD-P-355.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dropping pipets suitable for dispensing liquids.

1.2 Classification.

1.2.1 Types. Dropping pipets shall be of the following types as specified (see 6.1):

Type I - Glass.
Type II - Plastic, sterile.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

Federal Specifications

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.
PPP-B-676	Box, Setup.

FSC 6530

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Federal Standard

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications and standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, MO, Dallas, Denver, San Francisco, Los Angeles, Seattle and Washington, DC.

(Federal Government activities may obtain copies of Federal specifications and standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-S-36586	Sterilization Test Strip Set, Bacterial Spore.

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation C148	Polariscopic Examination of Glass Containers.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

U.S. PHARMACOPOEIAL CONVENTION PUBLICATION, INC.

Pharmacopoeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass. Glass shall be soda lime.

3.1.2 Rubber. Bulb shall be made of oil-resistant synthetic rubber.

3.1.3 Plastic. Plastic shall be polyethylene, polyvinyl chloride, cellulose acetate, or regenerated cellulose (cellophane).

3.2 Style and design.

3.2.1 Type I. Glass dropping pipet shall be that commonly known as Medicine Dropper, as described in the U.S.P., and shall be furnished complete with a bulb.

3.2.1.1 Pipet. The overall length of the pipet shall be 3 inches (in.) $\pm$ 1/8 in. The outside diameter of the unconstricted portion of the pipet, exclusive of the collar, shall be 1/4 in. $\pm$ 1/16 in. The outside diameter at its delivery end shall be 3 $\pm$ 0.5 millimeters (mm.). The upper end of the pipet shall be flanged to fit the inner ring in the bulb. The ends of the glass pipet shall be smoothly finished.

3.2.1.1.1 Strain. The finished pipet shall show no ring strain greater than Temper No. 1, and any longitudinal strain shall be faint and highly diffused when the pipet is tested as specified in 4.4.1.

3.2.1.2 Bulb. Bulb shall be constructed with heavy walls, shall fit tightly around the collar of the pipet, and shall be of sufficient drawing strength, when completely compressed and released, to fill at least two-thirds of the length of the pipet. Bulb shall be made with an inner ring to fit around the collar of the pipet so as to prevent the pipet from slipping up into the rubber bulb.

3.2.1.2.1 Drop dispensing. The glass pipet shall deliver 20 drops of water, the total weight of which is between 0.9 grams and 1.1 gram at 15°Centigrade (°C.), when tested as specified in 4.4.2.

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containers of the type, size, and kind com-

3.2.1.2.2 Bulb leakage. Bulb shall be made to fit the unconstricted end of the pipet, so that it shall pass the air test specified in 4.4.3 without showing leakage around the pipet or other defects in the rubber.

3.2.1.2.3 Oil resistance. Bulb shall be resistant to light mineral oil and corn oil and shall not exceed the following maxima, when tested as specified in 4.4.4.

Test media	Maximum increase by weight	
	72°F.	99°F.
Light mineral oil	8.3 percent	12.8 percent
Corn oil	5.1 percent	11.5 percent

3.2.1.2.4 Aging. The bulb shall be subjected to the accelerated aging test specified in 4.4.5. After 24 hours at room temperature, there shall be no change in the condition of the bulb that might adversely affect its serviceability.

3.2.2 Type II pipet. Plastic dropping pipet shall be a disposable, compressible pipet (dropper), of the type used for dispensing ophthalmic preparations. The pipet shall not exceed 4 in. in length and shall be closed at one end. The open end shall be cut smooth or rounded and shall not exceed 5/16 in. in outside diameter.

3.2.2.1 Sterility. Plastic pipet shall be sterile when tested as specified in 4.4.6.

3.2.2.1.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National Stock Number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy

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forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.2.2.1.2 Lot size. A sterility lot size shall consist of a manufacturer's batch subjected to the same sterilizer load. Each unit package shall be clearly marked with a lot number as specified in 5.3.2 which shall be traceable to the pertinent manufacturing batch and date of sterilization.

3.2.2.2 Envelope. An envelope or other suitable type container shall completely enclose each plastic pipet. The envelope shall be adequately sealed and shall maintain sterility of the pipet until manually opened (see 4.4.6). Closure or flap of the envelope shall be so constructed as to afford easy access to the enclosed pipet.

3.3 Workmanship. Workmanship shall be first class throughout. Pipet shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the bulb rubber and/or glass portion of pipet, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the name of the

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contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the components, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. The unit of product for examination shall be the individual pipet.

Table I Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. The unit of product for tests shall be the individual pipet except for the sterility test. Unit of product for the sterility test shall be a box. A specimen shall consist of one pipet. Four specimens shall be selected from each unit of product (or box) for the sterility test. Failure of one or more of the four specimens shall constitute failure of the unit of product and the entire lot.

Table II Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL
Type I				
Strain	3.2.1.1.1	4.4.1	S-1	*
Drop dispensing	3.2.1.2.1	4.4.2	S-1	*
Bulb leakage	3.2.1.2.2	4.4.3	S-1	*
Oil resistance	3.2.1.2.3	4.4.4	S-1	*
Aging	3.2.1.2.4	4.4.5	S-1	*
Type II				
Sterility	3.2.2.1	4.4.6	S-3 **	*

* Acceptance number shall be zero for all sample sizes.
** Minimum of 40 specimens.

4.3 Examination. The pipets shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with Table III.

Table III Classification of defects

Categories	Defects *
Major	
101	Glass or plastic pipet not free of cracks, pin holes or breaks.
102	Bulb for Type I pipet not free of cracks, holes or breaks.
103	Edges of pipet not free of sharpness.
104	Open ends of pipet not clear and free of obstructions.
Minor	
201	Glass pipet not free of bubbles, stones or striae which might affect serviceability.
202	Pipet not clean and free of dirt and foreign matter.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The pipet shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Strain. The glass pipet shall be tested for ring strain in accordance with Method A or B of ASTM Designation C148.

4.4.2 Drop dispensing test. With the pipet held in the vertical position, 20 drops shall be dispensed into a weighing bottle. Weight shall be determined accurate to the nearest 0.01 gram.

4.4.3 Bulb leakage. The pipet shall be subjected to 20 pounds per square inch (gage) internal air pressure.

4.4.4 Oil resistance test. Bulb shall be tested as follows:

a. Divide samples into four equal groups and accurately weigh each group.
b. Immerse two groups in separate containers of mineral oil.
c. Immerse two groups in separate containers of corn oil.
d. Allow one mineral oil container and one corn oil container to stand for 72 hours at 72°F. ± 2°.
e. Allow one mineral oil container and one corn oil container to stand for 72 hours at 99°F. ± 2°.

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f. Remove the bulb from the test media keeping the groups separate, place on a screen with the bulb opening facing down and allow to drain for 30 minutes.
g. Wipe bulb clean of oil with a towel and allow to stand at room temperature for 16 hours.
h. Reweigh and calculate percentage increase in weight.

4.4.5 Aging test. Test bulbs, at a temperature of 70°C. ± 1°, shall be subjected to the action of oxygen under 300 pounds pressure per square inch (gage) for 3 days.

4.4.6 Sterility test. The plastic dropping pipet (Type II) shall be tested for sterility in accordance with the method specified in the U.S.P. for non-parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.6.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II.

4.4.6.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of sterilization	Organism	Incubation temperature
Steam	B. stearothermophilus	55° - 65°C.
Dry heat	B. stearothermophilus, or	55° - 65°C.
	B. subtilis var. niger (or globigii)	30° - 35°C.
Gas	B. subtilis var. niger (or globigii)	30° - 35°C.
Radiation	B. pumilus	30° - 35°C.

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.6.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.6 and 4.4.6.1. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.6.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P.

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4.4.6.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the contracting officer, and approval obtained before employment of the alternate method on a Government contract.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV Examination of preparation for delivery

Examination	Defects
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One box, containing twelve pipets of Type I or one hundred pipets of Type II as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit package. Each unit of issue of Type I or Type II shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Type I pipets shall be arranged in such a manner as to prevent glass-to-glass contact. Closure of box shall be adequate to prevent accidental opening.

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5.1.1.1.2 Intermediate package. Twenty four units of issue of Type I shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, class domestic. Twelve units of issue of Type II shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, class domestic or PPP-B-676. Closure of boxes shall be adequate to prevent accidental opening.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Two hundred eighty eight units of issue of Type I or five hundred seventy six units of issue of Type II shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

5.2.2 Level B. Two hundred eighty eight units of issue of Type I or five hundred seventy six units of issue of Type II shall be packed in a box designed for a Type 2 load, and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2. Military activities.

5.3.2.1 Interior packages. Each Type II pipet, packaged as specified in 3.2.2.2, shall be commercially identified. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. In addition, for Type II pipet,

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marking shall include the method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry heat sterilization). Marking shall follow the level of packaging and date of packaging. (Example: "A-1/75-G.") Final sterilization shall be on or after indicated packaging date. Lot (control number shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, containers of Type II pipets shall be marked with method of sterilization as specified in 5.3.2.1 and lot (control) number.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type required (see 1.2.1).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6530-00-299-8631	PIPET, DROPPING, Plastic, Disposable, Sterile, 100s
6530-00-422-8120	PIPET, DROPPING, Glass, Rubber Bulb, 12s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6530-1339
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DOCUMENT IDENTIFIER AND TITLE
DD-P-00355A (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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